

# DECODABLE, NOT CASHABLE

A P R E P R I N T

AUXI & CAITLYN MEEKS

AuxiLab Tenerife  
research@auxi.cafe

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## ABSTRACT

**Abstract.** We built a reservoir designed for tidy, addressed storage — a card index — and raced it against the lab's standard mixing reservoir on the same text, same readouts, same everything. The card index stores history **strictly deeper** (its past can be decoded further back) and predicts the next character **strictly worse**, in all four seeds tested, at both spectral radii, 12/12 strict inequalities on replication. The gap grows with spectral radius: depth  $+0.9 \rightarrow +3.0$  characters, prediction cost  $+0.07 \rightarrow +0.15$  bits. A follow-up proves the failure is structural: handing the readout explicit linear "unbinding" keys to every drawer is a readout reparametrization and cannot help even in principle — the gap survives ( $0.081 \rightarrow 0.091$  at  $\rho=0.95$ ). A crossover sweep locates where the trade inverts: below  $\rho^* \approx 0.72$  the tidy delay line actually cashes better. Storing memories and using them are different talents, demonstrated on purpose.

**Keywords** reservoir computing · echo state networks · permutation matrices · holographic reduced representations · memory vs. prediction · pre-registration

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## §1 The question

The lab's standard machine is a "pond":  $N = 1024$  fixed random tanh neurons, sparse random recurrence, reading text one character at a time — a reservoir computer in the classical echo-state sense [1][2][3]. It never trains; only a small readout on top does. A pond *mixes* — every step smears the incoming character into a shared, entangled state.

What if it filed instead? Replace the mixing matrix with  $W = \rho \cdot P$ , where  $P$  is a single random 1024-cycle permutation: the state no longer mixes, it *rotates*, each step moving the whole memory one drawer along the cycle. Give the inputs Hadamard codes — 27 exactly orthogonal input columns (scaled  $1/\sqrt{3}$ ), one per character — so deposits don't collide: the permutation-and-superposition storage of holographic reduced representations [4] and hyperdimensional computing [5], built into the recurrence. That is the **addressed pond**: a filing cabinet where yesterday's character sits  $j$  drawers along, undisturbed. Intuition says organized storage should help. Does it?

Two rulers, and the whole finding lives in their disagreement:

- **Decode depth** (the U3 ruler): how many characters back a ridge probe can still read out of the state — the interpolated 50%-accuracy crossing over lags 1–32.

- **Cashing:** validation bits-per-character (bpc) of the house logistic readout predicting the *next* character. Lower is better. For scale on these splits: uniform guessing 4.7549, 3-gram 2.9279, 5-gram 2.1585.

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## §2 The machine

One registered  $2 \times 2 (\times 2)$  factorial, 24 cells: recurrence  $W \in \{\text{perm: } \rho \cdot P; \text{ sparse: fanin-10 random, rescaled to spectral radius } \rho\} \times \text{input coding } W_{\text{in}} \in \{\text{had: Hadamard; rnd: uniform random}\} \times \text{activation } \{\text{tanh, linear}\}$ , with tanh at  $\rho \in \{0.6, 0.95, 1.1\}$ . Data: 2M training / 500k validation / 500k test characters of text8 [6], washout 200. `perm_had` is the addressed pond; `sparse_rnd` is the standard control. One useful physical fact: a permutation is norm-preserving, so the addressed pond runs stably at  $\rho > 1$ , past the edge where mixing ponds normally collapse.

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## §3 What we registered in advance

Six ordinal predictions and four quantitative bands, committed before the run (wording finalized after a disclosed 100k-character smoke test — a disclosure the registration protocol requires). The headline bet, O6: the addressed pond would be *both* strictly deeper on decode *and* strictly worse on logistic bpc than the standard pond at  $\rho \in \{0.95, 1.1\}$  — the claim that storage is bought by giving up mixing, stated as a conjunction so that either half failing would break it.

Because O6 was a surprising claim, the house rules made it merely provisional until a registered replication on seeds 1–3: 6 cell-pairs, 12 strict inequalities, with the falsifiers named in writing before the rerun — *any* depth inequality breaking would demote the claim to an open question; any bpc inequality breaking (depth intact) would half-falsify it; both breaking would file seed 0 as a candidate outlier. 12/12 was the only outcome allowed to promote.

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## §4 What happened

Run #1 (seed 0): 6/6 ordinal predictions hit, 4/4 quantitative bands hit. The registered headline, in numbers:

**Table 1.** Addressed pond (`perm_had`, `tanh`) vs standard pond (`sparse_rnd`, `tanh`), seed 0. Depth in characters (higher = stores deeper); logistic val bpc (lower = predicts better).

| $\rho$ | depth <code>perm_had</code> | depth <code>sparse_rnd</code> | bpc <code>perm_had</code> | bpc <code>sparse_rnd</code> |
|--------|-----------------------------|-------------------------------|---------------------------|-----------------------------|
| 0.95   | 11.05                       | 10.11                         | 2.7538                    | 2.6727                      |
| 1.1    | 13.72                       | 10.63                         | 2.8779                    | 2.7337                      |

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Deeper on the left, worse on the right, at both radii — and the W-axis depth gap *grows* with  $\rho$  (roughly  $+0.4 \rightarrow +1.0 \rightarrow +3.2$  characters across  $\rho = 0.6 / 0.95 / 1.1$ ), with no edge-of-chaos collapse on the addressed arm. The replication, seeds 1–3:

**Table 2.** Registered replication (run ledger #2), seeds 1-3. Every registered inequality strict: 12/12.

| seed | $\rho$ | depth: perm > sparse | bpc: perm > sparse |
|------|--------|----------------------|--------------------|
| 1    | 0.95   | 11.04 > 10.10        | 2.7613 > 2.6681    |
| 1    | 1.1    | 13.70 > 10.64        | 2.8833 > 2.7303    |
| 2    | 0.95   | 11.02 > 10.12        | 2.7480 > 2.6751    |
| 2    | 1.1    | 13.68 > 10.70        | 2.8699 > 2.7360    |
| 3    | 0.95   | 11.03 > 10.07        | 2.7565 > 2.6671    |
| 3    | 1.1    | 13.67 > 10.63        | 2.8744 > 2.7209    |

The seed-to-seed spread is unexpectedly small: across four seeds the addressed pond's depth spans 11.02–11.05 at  $\rho=0.95$  and 13.67–13.72 at 1.1 — a spread of  $\leq 0.05$  characters, about half a percent of the gap being tested. Plausibly because a random N-cycle is the same object up to relabeling, the depth of an addressed pond is essentially deterministic in the seed.

**Can the depth be cashed with a key?** A natural objection: perhaps the readout simply cannot *reach* the drawers. We therefore handed it the keys explicitly — features  $[x \parallel P^{-1}x \parallel \dots \parallel P^{-4}x]$ , the state unbound, in the holographic-representation sense [4], by the pond's own cycle. The registered analysis proved this vacuous before the run: each appended block is a fixed coordinate permutation of  $x$ , so the widened features span *exactly* the raw state's linear function class — a linear readout already owns every fixed permutation of its input. The run confirmed the theorem's observable edge: ridge validation identical to four decimals in every arm, and the perm–sparse logistic gap *survived*,  $0.0811 \rightarrow 0.0906$  at  $\rho=0.95$  and  $0.1443 \rightarrow 0.1484$  at  $\rho=1.1$ . Linear unbinding cannot recover the stored depth even in principle. (A nonlinear drawer-opener is a named, not-yet-run follow-up.)

**Where the trade inverts.** A registered low- $\rho$  sweep (seed 0, random input coding both arms, isolating the W axis) measured the gap ladder,  $\text{gap}(\rho) = \text{sparse} - \text{perm}$  logistic val bpc (positive = the delay line wins):

**Table 3.** The mixing-tax crossover (run ledger #6, seed 0, single-seed measured context).

| $\rho$            | perm_rnd bpc | sparse_rnd bpc | gap     | winner     |
|-------------------|--------------|----------------|---------|------------|
| 0.3               | 2.8969       | 2.9028         | +0.0059 | delay line |
| 0.45              | 2.7758       | 2.7865         | +0.0107 | delay line |
| 0.6               | 2.6843       | 2.6989         | +0.0146 | delay line |
| $\rho^* = 0.7227$ | —            | —              | 0       | crossover  |
| 0.8               | 2.6611       | 2.6519         | −0.0092 | mixing     |
| 0.95              | 2.7353       | 2.6727         | −0.0626 | mixing     |

Below  $\rho^* \approx 0.72$  the pure delay line cashes *better*; above it, mixing wins and the tax grows. Storage ordering — the addressed pond decodes deeper — held at every  $\rho$  tested, 0.3 through 1.1. Both architectures have their bpc optimum near  $\rho=0.8$ , just above the crossover: at the operating point

where these ponds predict best, filing and mixing are nearly equivalent, and they only differentiate away from it — mixing degrades gracefully, the delay line collapses.

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## §5 What it means

Storing memories neatly and using them are different talents, and here the trade is built on purpose: the permutation buys addressable, collision-free storage by giving up nonlinear mixing, and the readout pays for the missing mixing in bits. The wall is structural, not a missing tool — even the key to every drawer doesn't help, because "apply a fixed permutation" is something a linear reader could always do for itself. What the prediction readout wants is not deep, tidy, decodable storage; it wants the computed, entangled summaries that mixing provides. Decodable  $\neq$  cashable.

The one honest escape hatch left open: only *linear* unbinding is ruled out. Whether any cheap *nonlinear* reader can cash the stored depth is a named open question in the lab's queue. Until then, the drawers stay shut.

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## §6 References

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2. W. Maass, T. Natschläger, H. Markram, "Real-time computing without stable states: a new framework for neural computation based on perturbations," *Neural Computation* 14(11), 2002.
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5. P. Kanerva, "Hyperdimensional computing: an introduction to computing in distributed representation with high-dimensional random vectors," *Cognitive Computation* 1(2), 2009.
6. M. Mahoney, "Large text compression benchmark" (text8), [matmahoney.net/dc/textdata](http://matmahoney.net/dc/textdata).

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## §7 Provenance

- **169b7c2d** — 2026-08-10-addressed-pond: main factorial, 24 cells, seed 0 (run ledger #1). Registered before running; all six ordinals and four bands hit.
- **866d975e** — 2026-08-11-addressed-seeds-replication: seeds 1–3, 12 cells (run ledger #2). 12/12 strict inequalities; promotion by the pre-stated rule.
- **472d4fb7** — 2026-08-11-addressed-unbinding-readout: the linear key-ring test, seed 0 (run ledger #5). Null registered in advance and confirmed; axis closed.
- **4052cdec** — 2026-08-11-addressed-low-rho-crossover: the crossover sweep, seed 0 (run ledger #6). Single-seed measured context, labeled as such.

All experiments registered before running, with named falsifiers. Headline claim tested on 4 seeds (0–3); replicated before publication. Unbinding and crossover results are single-seed and presented as supporting structure, not as the promoted claim.

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AuxiLab · findings are pre-registered and replicated before publication

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